SIDHARTH J

Final-Year Electrical & Electronics Engineering Student

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PROFESSIONAL SUMMARY

Final-year Electrical and Electronics Engineering student with proven expertise in microcontroller-based automation and PCB design. Delivered automated solar-tracking and street-lighting systems, achieving up to 35% higher energy efficiency and 30% lower energy consumption. Proficient in circuit design, MATLAB/Simulink, and PCB layout, with research published at an IEEE conference on green hydrogen production. Equipped to deliver high-impact electrical engineering solutions from day one.

EDUCATION

B.E. - Electrical and Electronics Engineering *2022 - 2026 VSB College of Engineering, Coimbatore | CGPA: 7.59*

Diploma - Electrical and Electronics Engineering *2020 - 2023 Sree Narayana Guru Polytechnic College | Percentage: 74.8%*

SSLC *2020 Sri P. Mallaiyan Matriculation Hr. Sec. School | Percentage: 63.8%*

PROJECTS

Dual-Axis Solar Tracking System *Smart Systems & Automation*

- Engineered a dual-axis solar tracking system using Arduino Nano, LDR sensors, and servo motors, improving energy capture efficiency by approximately 35%.
- Implemented microcontroller-based servo control logic with real-time sensor feedback, enabling continuous automated panel alignment without manual adjustment.
- Took ownership of hardware and circuit assembly, coordinating with 2 teammates on integration, testing, and troubleshooting to deliver a working prototype within 2 weeks.

Tech stack: Embedded C, Arduino, Servo Motors, LDR Sensors

Solar Street Light System *Arduino-Based Automation*

- Designed and programmed an automatic solar street lighting system using LDR sensors and MOSFET switching logic, cutting energy consumption by approximately 30%.
- Applied microcontroller programming principles to achieve 100% automated switching control and real-time light detection, eliminating manual operation entirely.
- Tested system performance under varying ambient light conditions to ensure reliable day/night switching accuracy.

Tech stack: Embedded C, Arduino, LDR Sensors, MOSFETs

INTERNSHIP EXPERIENCE

Embedded Systems Intern - Study Shinee Software Solution *Dec 2025 - Jan 2026*

- Developed 4+ microcontroller-based modules for firmware programming and sensor integration, enabling accurate real-time data acquisition for IoT prototypes.
- Built hardware-software integration across 4+ IoT-based prototypes using C on ARM-based platforms
- Applied 2 communication protocols (UART, I2C) in firmware development, ensuring reliable, low-latency sensor-to microcontroller data transfer.

PCB Design Training Intern - A.D. Engineering Solution *June 2023*

- Designed and reviewed 4+ PCB layouts, performing component routing, schematic design, and circuit validation for microcontroller-based hardware prototypes.
- Applied PCB fabrication standards and design rules, reducing layout errors and improving manufacturability of hardware prototypes.

SKILLS

Electrical & Circuit Design: Circuit Analysis, Network Theory, PCB Design, Circuit Layout, Component Routing, Schematic Design

Engineering Software: MATLAB/Simulink, AutoCAD Electrical, Proteus, Arduino IDE,KiCad

Programming & Firmware: C, Arduino IDE

Hardware & Components: LDR Sensors, Servo Motors, MOSFETs

Core Electrical Concepts: Power Electronics, Control Systems, Renewable Energy Systems, Automation, Energy Efficiency Analysis

PUBLICATION

THERMO-LYTIC: Multi-Source Hybrid for Green Hydrogen Production | *IEEE Conference Paper (2025)*

● Authored and presented a paper on green energy, hydrogen production, hybrid systems, and thermolysis at an IEEE conference.

CERTIFICATIONS

● NPTEL Certification - Internet of Things (IoT), IIT Kharagpur

AREAS OF INTEREST

Power Electronics, Microcontrollers & IoT, Circuit Analysis & Network Theory, PCB Design & Hardware Development, Automation & Control Systems, Renewable Energy & Solar Systems

LANGUAGES

Tamil, English